

Multi-VLAN Office Network Project

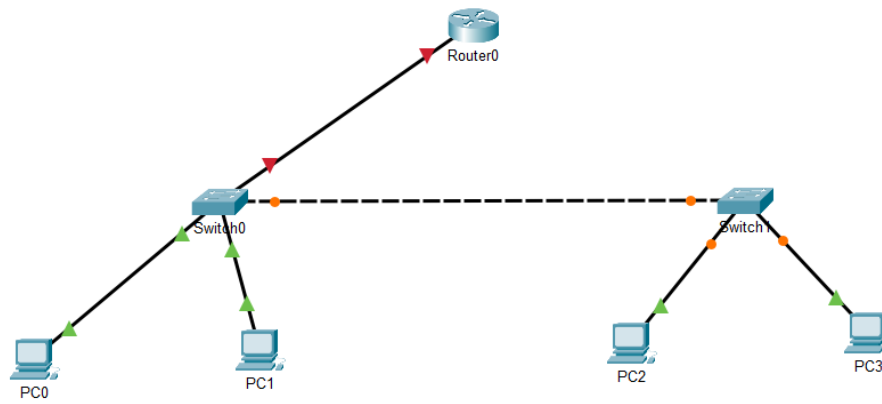
Inter-VLAN routing, port security, and ACLs — Cisco Packet Tracer

Prepared by: Faraz

Section 1: Project Evidence

Network Topology

Router0 - Switch0 - Switch1, with IT/Sales on Switch0 and Finance/Servers on Switch1.



Switch0 - VLANs and Trunk Configuration

show vlan brief and show interfaces trunk on Switch0. VLANs 10 (IT) and 20 (Sales) assigned; trunk ports carrying all VLANs to Router0 and Switch1.

```
SW0#sh vlan brief
VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   IT                     active    Fa0/1
20   Sales                  active    Fa0/2
30   Finance                active
40   Servers                active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

SW0#sh int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/3     on        802.1q         trunking    1
Fa0/4     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/3     1,10,20,30,40
Fa0/4     1,10,20,30,40

Port      Vlans allowed and active in management domain
Fa0/3     1,10,20,30,40
Fa0/4     1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/3     1,10,20,30,40
```

Switch1 - VLANs and Trunk Configuration

show vlan brief and show interfaces trunk on Switch1. VLANs 30 (Finance) and 40 (Servers) assigned; trunk port to Switch1 confirmed.

```
SW1#sh vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	IT	active	
20	Sales	active	
30	Finance	active	Fa0/1
40	Servers	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1#sh int trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/4	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Fa0/4	1,10,20,30,40			
Port	Vlans allowed and active in management domain			
Fa0/4	1,10,20,30,40			
Port	Vlans in spanning tree forwarding state and not pruned			
Fa0/4	1,10,20,30,40			

Router0 - Sub-interfaces (Router-on-a-Stick)

show ip interface brief confirming all 4 sub-interfaces (gi0/0.10, .20, .30, .40) are up/up with correct gateway IPs.

```
R1#sh ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset    up          up
GigabitEthernet0/0.10 192.168.100.1   YES manual   up          up
GigabitEthernet0/0.20 192.168.100.65  YES manual   up          up
GigabitEthernet0/0.30 192.168.100.129 YES manual   up          up
GigabitEthernet0/0.40 192.168.100.193 YES manual   up          up
GigabitEthernet0/1    unassigned      YES unset    administratively down down
GigabitEthernet0/2    unassigned      YES unset    administratively down down
Vlan1             unassigned      YES unset    administratively down down
```

Connectivity Test: IT to Finance

PC0 (192.168.100.2, VLAN 10) successfully pings PC2 (192.168.100.130, VLAN 30) - 0% packet loss.

```
C:\>ping 192.168.100.130
```

```
Pinging 192.168.100.130 with 32 bytes of data:
```

```
Reply from 192.168.100.130: bytes=32 time=1ms TTL=127
```

```
Reply from 192.168.100.130: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.100.130: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.100.130: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 192.168.100.130:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Connectivity Test: IT to Servers

PC0 (VLAN 10) successfully pings PC3 (192.168.100.194, VLAN 40) - 0% packet loss.

```
C:\>ping 192.168.100.194
```

```
Pinging 192.168.100.194 with 32 bytes of data:
```

```
Reply from 192.168.100.194: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.100.194: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.100.194: bytes=32 time=1ms TTL=127
```

```
Reply from 192.168.100.194: bytes=32 time=1ms TTL=127
```

```
Ping statistics for 192.168.100.194:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Connectivity Test: Finance to Servers

PC2 (VLAN 30) successfully pings PC3 (VLAN 40) - 0% packet loss.

```
C:\>ping 192.168.100.194
```

```
Pinging 192.168.100.194 with 32 bytes of data:
```

```
Reply from 192.168.100.194: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.100.194: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.100.194: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.100.194: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 192.168.100.194:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```


ACL Test: Sales to Servers (Blocked as Designed)

PC1 (Sales, VLAN 20) attempts to ping PC3 (Servers, VLAN 40). Extended ACL 100 blocks this traffic; Router0 replies with Destination host unreachable from its own Sales gateway interface (192.168.100.65), confirming the ACL is working correctly.

```
C:\>ping 192.168.100.194

Pinging 192.168.100.194 with 32 bytes of data:

Reply from 192.168.100.65: Destination host unreachable.
Reply from 192.168.100.65: Destination host unreachable.
Reply from 192.168.100.65: Destination host unreachable.
Reply from 192.168.100.65: Destination host unreachable.

Ping statistics for 192.168.100.194:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Switch0 - Port Security (Before/After Learning MAC)

show port-security interface fa0/1 on Switch0, before and after traffic causes the sticky MAC address to be learned.

```
SW0#sh port-security int fa0/1
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 0
Configured MAC Addresses : 0
Sticky MAC Addresses    : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```

```
SW0#sh port-security int fa0/1
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 0001.C7B5.00E1:10
Security Violation Count : 0
```

Switch1 - Port Security

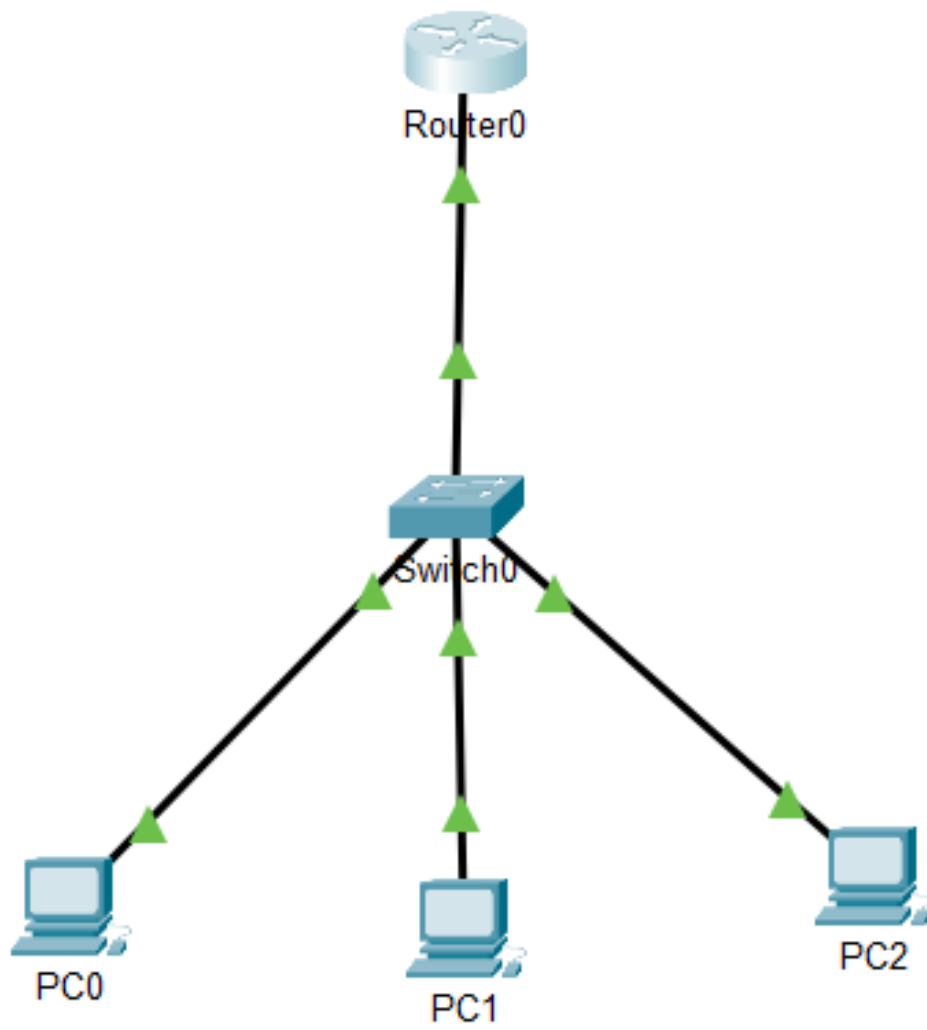
show port-security interface fa0/1 on Switch1, confirming sticky MAC address learned and violation mode set to shutdown.

```
SW1#show port-security interface fa0/1
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 000A.41D5.0A3B:30
Security Violation Count : 0
```

Section 2: Appendix - Earlier Practice

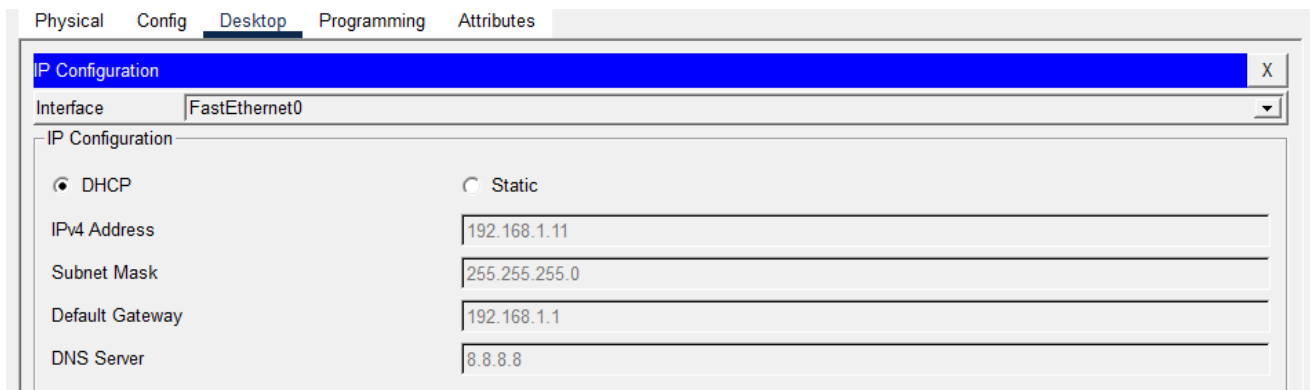
Appendix: Day 1 - ipconfig Practice

Early practice task, not part of this project.



Appendix: Day 2 - ipconfig /all Practice

Early practice task, not part of this project.



The screenshot shows a network configuration window titled "IP Configuration" with a blue header bar and a close button (X) on the right. Below the header, there is a tabbed interface with four tabs: "Physical", "Config", "Desktop", and "Attributes". The "Config" tab is selected. Under the "Config" tab, there is a dropdown menu labeled "Interface" showing "FastEthernet0". Below this, there is a section titled "IP Configuration" with two radio buttons: "DHCP" (selected) and "Static". Below the radio buttons, there are four text input fields for the following parameters:

Parameter	Value
IPv4 Address	192.168.1.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8

Appendix: Day 3 - Static Routing Lab (Config)

Early practice task, not part of this project.

10	IT	active	Fa0/24, Gig0/1, Gig0/2
20	Sales	active	Fa0/1
			Fa0/2

Appendix: Day 3 - Static Routing Lab (Output)

Early practice task, not part of this project.

```
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```


Appendix: Day 3 - Trunk Output Practice

Early practice task, not part of this project.

```
Switch#sh int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/3     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/3     1-1005

Port      Vlans allowed and active in management domain
Fa0/3     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/3     1 10 20
```

Appendix: DNS/Ping Practice

Supplementary ping practice.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.100.166

Pinging 192.168.100.166 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.100.166:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.100.130

Pinging 192.168.100.130 with 32 bytes of data:

Request timed out.
Reply from 192.168.100.130: bytes=32 time=12ms TTL=127
Reply from 192.168.100.130: bytes=32 time<1ms TTL=127
Reply from 192.168.100.130: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.100.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 4ms

C:\>ping 192.168.100.194

Pinging 192.168.100.194 with 32 bytes of data:

Request timed out.
Reply from 192.168.100.194: bytes=32 time<1ms TTL=127
Reply from 192.168.100.194: bytes=32 time=1ms TTL=127
Reply from 192.168.100.194: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.100.194:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Appendix: Switch0 Port Security (Initial Check)

Early port-security verification before MAC was learned.

```
SW0#sh port-security int fa0/1
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 0
Configured MAC Addresses : 0
Sticky MAC Addresses    : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```